

The Mathematics of SVM: Hard Margin

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1. Overview

Key point: We write the SVM line as $w^T x + b = 0$ and its two edges as $w^T x + b = +1$ and -1 . The margin between the edges is then $2/\|w\|$, and SVM maximises it while every point stays on its correct side.

The previous Note, the [SVM intuition Note](#), described SVM in pictures: among all lines that separate two classes, choose the one whose margin d , the distance between the positive hyperplane π^+ and the negative hyperplane π^- , is the largest. This Note turns that picture into a formula that a computer can optimise.

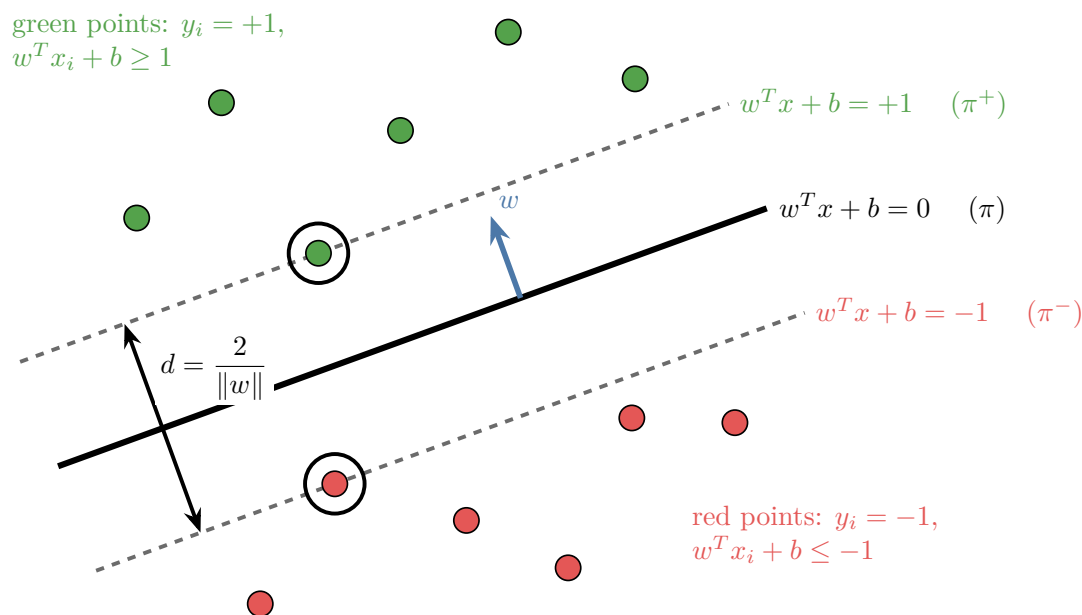


Figure 1: The three hyperplanes of an SVM, their equations, and the margin

Figure 1 shows the result. The steps to get there:

1. A **decision rule** that says on which side of the line a new point lies.
2. Equations for π^+ and π^- , and why they use $+1$ and -1 .
3. **Constraints** that keep every training point on its correct side.
4. A formula for the margin d in terms of w .
5. The optimisation problem that puts it all together.

2. The decision rule

Key point: Predict $+1$ if $w \cdot u + b \geq 0$, and -1 if $w \cdot u + b < 0$.

Before finding the best line, we need to know how a line classifies a new point once we have it.

2.1 A vector perpendicular to the hyperplanes

Key point: π , π^+ and π^- are parallel, so one vector w is perpendicular to all three. At first we know its direction but not its length.

The three hyperplanes are parallel. So a vector w that is perpendicular to one of them is perpendicular to all three (the blue arrow in Figure 1). At this point we only fix the **direction** of w ; its length is still unknown and will be decided later.

2.2 Projecting a new point onto w

Key point: Project the new point u onto w . If the projection passes a threshold c , the point is positive.

Take an unknown point u . In 2D a point is also a vector from the origin, with an x and a y component. Drop u onto the direction of w : this is the **projection** of u onto w , and its length grows with the **dot product** $w \cdot u$ (Figure 2).

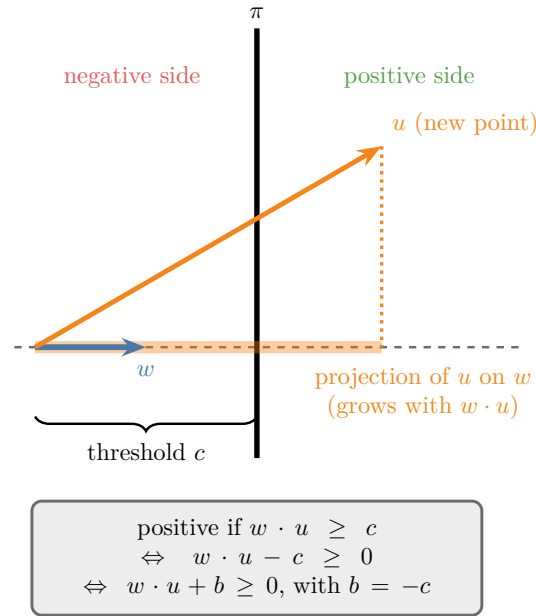


Figure 2: The decision rule: a point is positive when its projection onto w passes the threshold c

The boundary π crosses the direction of w at some unknown distance c from the origin. A point is on the positive side when its projection goes past that threshold:

$$w \cdot u \geq c \quad \Leftrightarrow \quad w \cdot u - c \geq 0 \quad \Leftrightarrow \quad w \cdot u + b \geq 0, \quad \text{with } b = -c$$

So the **decision rule** of SVM is:

$$\hat{y} = \begin{cases} +1 & \text{if } w \cdot u + b \geq 0 \\ -1 & \text{if } w \cdot u + b < 0 \end{cases}$$

Once we know w and b , classifying any new point takes one dot product and one addition.

Extra: Strictly, the projection's length is $w \cdot u / \|w\|$, where $\|w\|$ is the length of w . Comparing it with c gives $w \cdot u \geq c \|w\|$. That is the same rule with $b = -c \|w\|$, so the final form does not change.

2.3 The same rule in 2D

Key point: In 2D, $w \cdot u + b$ is just “put the point into the line's equation”, as in the perceptron trick Note.

In 2D this rule is the side test of the [perceptron trick Note](#): put the point into the left-hand side of the line's equation and look at the sign. Written with vectors, the line $2x + 3y + 3 = 0$ has $w = (2, 3)$ and $b = 3$.

- $u = (3, 2)$: $w \cdot u + b = 2 \times 3 + 3 \times 2 + 3 = 15 \geq 0$, so the point is **positive**.
- $u = (-2, 0)$: $w \cdot u + b = 2 \times (-2) + 3 \times 0 + 3 = -1 < 0$, so the point is **negative**.

The vector form says exactly the same thing, but it works unchanged in any number of dimensions.

3. The equations of π^+ and π^-

Key point: We choose π^+ to be $w^T x + b = +1$ and π^- to be $w^T x + b = -1$. Equal sizes keep π in the middle; the value 1 is a free choice of scale.

3.1 The assumption

Key point: π : $w^T x + b = 0$, π^+ : $w^T x + b = +1$, π^- : $w^T x + b = -1$.

To compute the margin we need equations for π^+ and π^- . We **choose** them to be:

$$\pi^+ : w^T x + b = +1, \quad \pi : w^T x + b = 0, \quad \pi^- : w^T x + b = -1$$

For the line $2x + 3y + 3 = 0$, these are $2x + 3y + 3 = 1$ and $2x + 3y + 3 = -1$: two lines parallel to it, one on each side. This choice raises three questions.

1. **Why must the two right-hand sides have the same size** (for example not $+1$ and -2)? Because π must lie exactly in the middle of π^+ and π^- . The distance from π to π^+ must equal the distance from π to π^- , and equal values on both sides give that.

2. **Why 1, and not 25 or 37?** The exact number makes no difference to which line is best. Section 5 shows that a different number only multiplies the margin by a constant, so 1 is chosen for convenience.
3. **Why are these equations useful at all?** Section 3.2 answers this with an experiment.

3.2 Scaling w and b changes the margin

Key point: Multiplying w and b by 10 gives the same line π , but π^+ and π^- move closer together. Dividing by 10 pushes them apart.

Multiply every coefficient of $2x + 3y + 3 = 0$ by 10. The result, $20x + 30y + 30 = 0$, is the **same line**: the slope and intercept do not change.

Now do the same to π^+ and π^- , multiplying only the left-hand side. The equations $20x + 30y + 30 = \pm 1$ are **not** the same lines as $2x + 3y + 3 = \pm 1$. Since the right-hand side stays at 1, the new π^+ and π^- lie much closer to π (Figure 3).

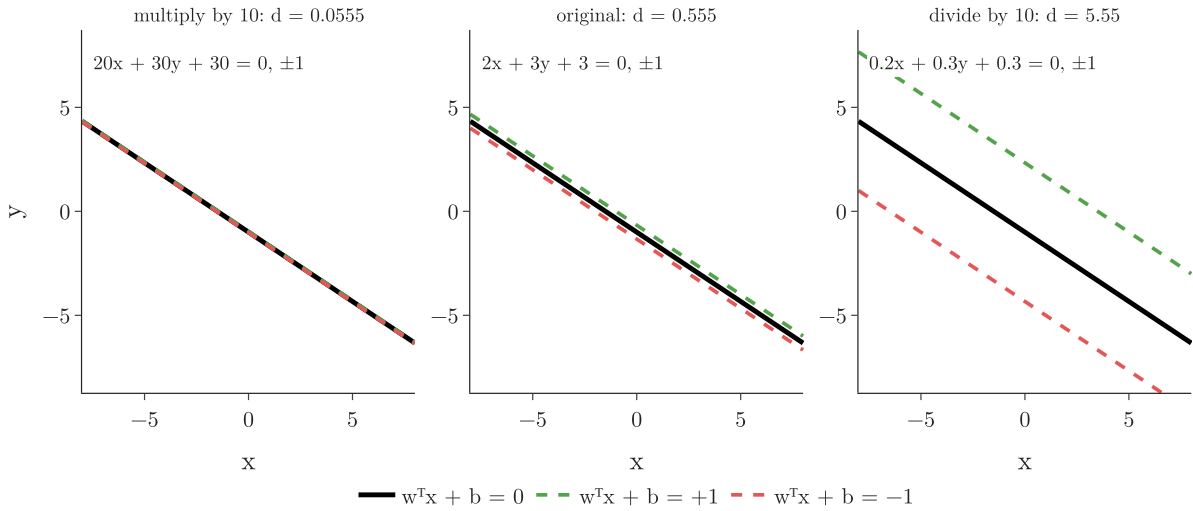


Figure 3: The same line π with w and b multiplied by 10, unchanged, and divided by 10

In plain words: scaling w and b up squeezes the margin; scaling them down widens it. As a formula (derived in Section 5), with $\|w\|$ the length of w :

$$d = \frac{2}{\|w\|}$$

With numbers, for $w = k \times (2, 3)$:

Factor k	Equation of π	$\ w\ $	Margin d
10	$20x + 30y + 30 = 0$	36.06	0.0555
1	$2x + 3y + 3 = 0$	3.606	0.555
0.1	$0.2x + 0.3y + 0.3 = 0$	0.3606	5.55

This is why the ± 1 equations are useful. While an optimiser adjusts w and b , it can both turn and shift the line and make the margin wider or narrower. The search for the best line becomes a search over w and b alone.

4. The constraints

Key point: Every training point must lie on or beyond its own hyperplane: $y_i(w^T x_i + b) \geq 1$ for every i .

4.1 Green points above π^+ , red points below π^-

Key point: Green: $w^T x + b \geq 1$. Red: $w^T x + b \leq -1$. Support vectors give exactly ± 1 .

The margin only means something if no point sits inside it. π^+ and π^- stop at the first point of each class, so no green point can lie below π^+ and no red point above π^- . In equations:

- every green point: $w^T x_i + b \geq 1$;
- every red point: $w^T x_i + b \leq -1$.

The support vectors lie exactly on π^+ or π^- , so they give exactly $+1$ or -1 . All other points give more than $+1$ or less than -1 .

4.2 One constraint for both classes

Key point: Label green points $+1$ and red points -1 , and multiply: both rules become $y_i(w^T x_i + b) \geq 1$.

Two separate rules are awkward to work with, so we merge them. Give every green point the label $y_i = +1$ and every red point the label $y_i = -1$, then multiply $w^T x_i + b$ by the label.

- **Green point:** $y_i = +1$ leaves $w^T x_i + b \geq 1$ unchanged.
- **Red point:** multiplying $w^T x_i + b \leq -1$ by -1 flips the sign: $-(w^T x_i + b) \geq 1$.

Both become one **constraint**, for every training point $i = 1, \dots, n$:

$$y_i(w^T x_i + b) \geq 1$$

with equality for the support vectors. Figure 4 checks this on the data of the SVM intuition Note, whose best line is $w = (0.049, 0.898)$, $b = -4.485$. For the red point $(2.5, 3)$: $w^T x + b = 0.049 \times 2.5 + 0.898 \times 3 - 4.485 = -1.67$, and $y_i(w^T x_i + b) = (-1)(-1.67) = 1.67 \geq 1$.

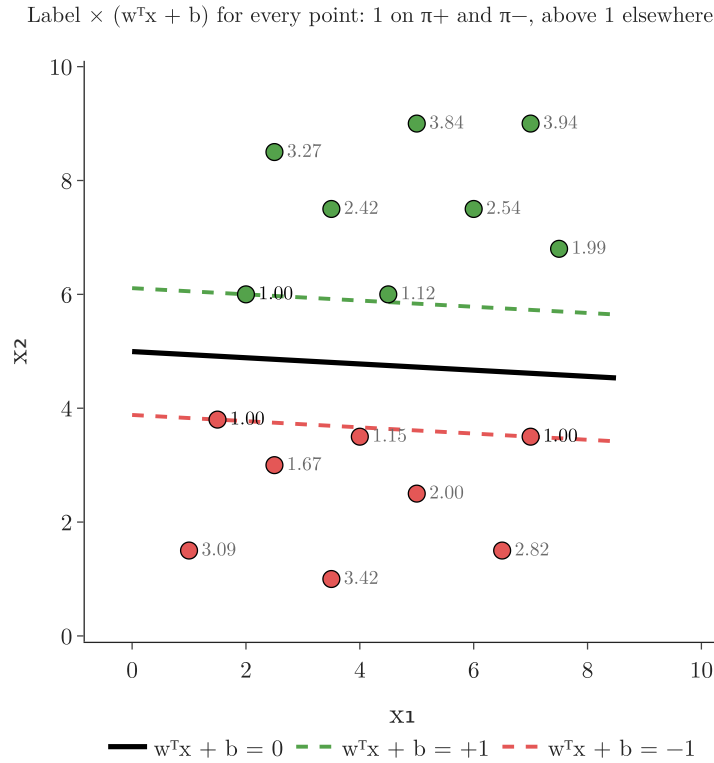


Figure 4: $y_i(w^T x_i + b)$ for every training point: exactly 1 for the three support vectors, more than 1 for the rest

The whole formula for the margin relies on these constraints. If a red point entered the region above π^- , or a green point the region below π^+ , the margin would no longer be the gap between the classes.

5. The margin in terms of w

Key point: Project the vector between two support vectors onto the unit vector $w/\|w\|$. The result is $d = 2/\|w\|$.

5.1 The derivation

Key point: $d = (x_2 - x_1) \cdot w/\|w\|$, and the equations of π^+ and π^- turn this into $2/\|w\|$.

Take one support vector on each edge: x_1 on π^- and x_2 on π^+ (Figure 5). The vector from x_1 to x_2 is $x_2 - x_1$. It crosses the margin, but at a slant, so its length is not the margin.

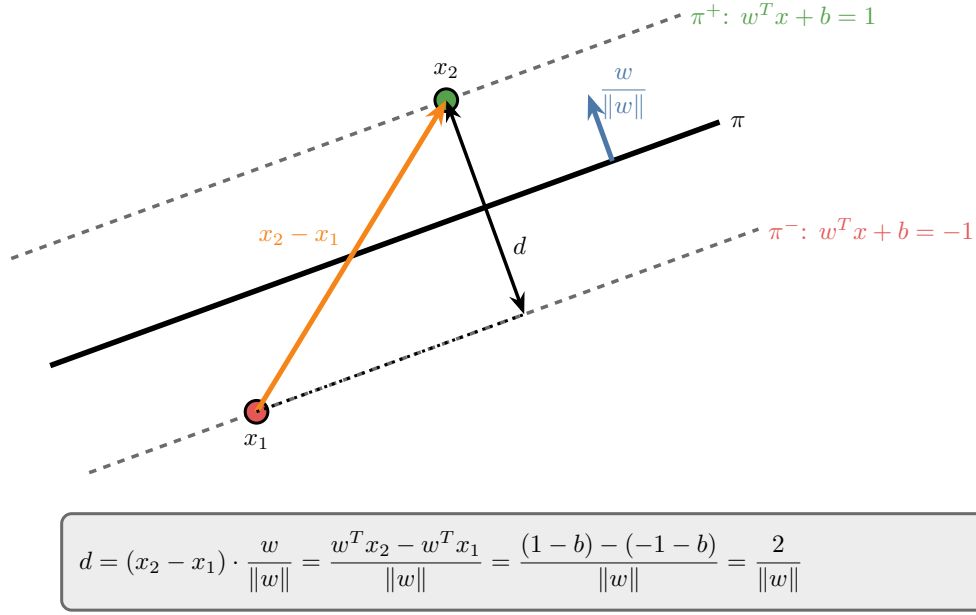


Figure 5: The margin is the projection of $x_2 - x_1$ onto the unit vector $w/\|w\|$

The **shortest** distance between the two edges is measured perpendicular to them, that is along w . So we project $x_2 - x_1$ onto the **unit vector** in the direction of w , which is w divided by its length (its **norm**) $\|w\|$:

$$d = (x_2 - x_1) \cdot \frac{w}{\|w\|} = \frac{w^T x_2 - w^T x_1}{\|w\|}$$

Both points are support vectors, so their equations are known:

- x_2 is on π^+ : $w^T x_2 + b = 1$, so $w^T x_2 = 1 - b$.
- x_1 is on π^- : $w^T x_1 + b = -1$, so $w^T x_1 = -1 - b$.

Substituting:

$$d = \frac{(1 - b) - (-1 - b)}{\|w\|} = \frac{2}{\|w\|}$$

5.2 The formula with numbers

Key point: For the best line of the intuition Note, $\|w\| = 0.899$ and $d = 2/0.899 = 2.22$.

The margin is two divided by the length of w . For the line of Figure 4, $w = (0.049, 0.898)$:

$$\|w\| = \sqrt{0.049^2 + 0.898^2} = 0.899, \quad d = \frac{2}{0.899} = 2.22$$

This is the margin $d = 2.22$ measured in the SVM intuition Note. The b cancelled out: the margin depends only on w . It is the full width between π^+ and π^- ; the one-sided distance from π to the nearest point, called the margin in the [perceptron code Note](#), is half of it, $1/\|w\|$.

Extra: This also answers “why 1?”. With $\pi^\pm : w^T x + b = \pm k$, the same steps give $d = 2k/\|w\|$. That is the old margin times a constant, so the same w and b are best. Equivalently, dividing w and b by k turns $\pm k$ back into ± 1 without moving any line. Fixing the support vectors at exactly ± 1 simply picks one scale for w and b .

6. The optimisation problem

Key point: Find w and b that maximise $2/\|w\|$, subject to $y_i(w^T x_i + b) \geq 1$ for every point.

Putting the pieces together, SVM looks for the w and b that make the margin as large as possible while every point respects its constraint:

$$w^*, b^* = \arg \max_{w, b} \frac{2}{\|w\|} \quad \text{such that} \quad y_i (w^T x_i + b) \geq 1 \quad \text{for all } i$$

This is a **constrained optimisation** problem: we maximise a function while keeping a condition true, one condition per training point. On the data of Figure 4 the answer is $w = (0.049, 0.898)$, $b = -4.485$ and $d = 2.22$.

Python: scikit-learn solves this problem for us. A very large C makes SVC behave like the hard-margin SVM.

```
import numpy as np
from sklearn.svm import SVC

svm = SVC(kernel="linear", C=1e6).fit(X, y)
w, b = svm.coef_[0], svm.intercept_[0]
y * (X @ w + b)           # every value >= 1
2 / np.linalg.norm(w)      # the margin: 2.22
```

The Notebook of this Note checks every number in this Note and rebuilds the scaling experiment of Figure 3 with a slider.

7. The limit: hard-margin SVM

Key point: The formulation needs perfectly separable data. A single point on the wrong side makes the constraints impossible to meet.

Real data is rarely perfectly separable. Usually it is **almost** linearly separable, with a few points on the wrong side, or not linearly separable at all.

The constraints above allow no exceptions. In Figure 6 one green point lies among the red

points: for the old line it gives $y(w^T x + b) = -2.47$, far below 1. No straight line can put that point on the green side without putting red points there too, so **no** w and b satisfy every constraint. The optimisation has no answer at all.

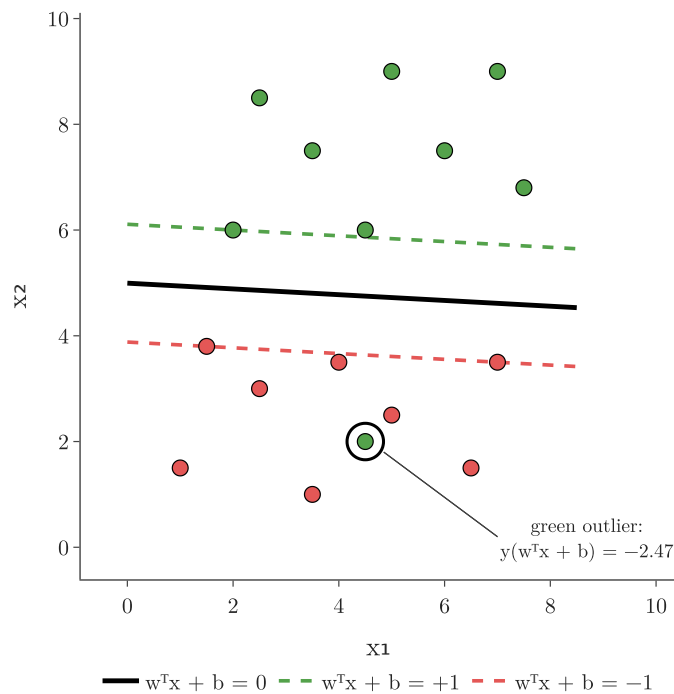


Figure 6: A single green outlier among the red points breaks the constraints of every line

The formulation of this Note is called the **hard-margin SVM**: it works only on data that a hyperplane separates perfectly. The next Note relaxes the constraints to allow a few mistakes, which gives the **soft-margin SVM**.

8. Summary

Piece	Formula
Decision rule	$\hat{y} = +1$ if $w \cdot u + b \geq 0$, else -1
Hyperplanes	$\pi : w^T x + b = 0, \pi^\pm : w^T x + b = \pm 1$
Constraint (every point)	$y_i(w^T x_i + b) \geq 1$, equal to 1 for support vectors
Margin	$d = 2/\ w\ $
Optimisation	maximise $2/\ w\ $ subject to the constraints

- w is perpendicular to all three hyperplanes; b shifts them.
- Scaling w and b leaves π in place but moves π^+ and π^- : a smaller $\|w\|$ means a wider margin.
- The labels ± 1 merge the two class rules into one constraint.
- Hard-margin SVM fails as soon as one point is on the wrong side.

9. Key terms

Term	Meaning
Decision rule (SVM)	Predict $+1$ if $w \cdot u + b \geq 0$ and -1 otherwise
Norm of a vector	The length of a vector, $\ w\ = \sqrt{w_1^2 + w_2^2 + \dots}$
Constraint	A condition the solution must satisfy; in SVM, $y_i(w^T x_i + b) \geq 1$ for every training point
Constrained optimisation	Maximising or minimising a function while keeping one or more constraints true
Hard-margin SVM	The SVM that allows no point inside the margin or on the wrong side; it needs perfectly separable data